

Many system-level investigations begin with two fundamental questions. Why did the system behave as it did? What would have happened had we intervened differently?

In the case of power systems, the answers to these questions depend on perspective. Economists often approach power-grid-related questions from the behavioral side, using observed real-world data to estimate how consumers, firms, and other actors respond to market signals and policy changes. Power and energy systems researchers, by contrast, rely on mathematical models to simulate how the system reacts under alternative technical conditions. Yet neither approach, on its own, captures the full picture. Econometric models can reveal real-world behavior, but they often miss the physical and operational constraints of transmission and distribution networks. Power and energy systems models can accurately represent physical and reliability limits, but they often rely on behavioral assumptions that are detached from real-world observations. When those assumptions are inaccurate, the models risk misestimating true system behavior and, ultimately, the long-term value of both engineering and policy interventions.

My research addresses this gap by linking empirically observed actor behavior with power-system models. I first use empirical energy-economics methods to estimate how electricity-sector actors respond to electricity prices, policy incentives, and power-system conditions. I then incorporate these behavioral responses into power-system models to evaluate how they affect system-level outcomes such as grid planning, operations, and market performance. In this way, my work creates a feedback loop between observed behavior and system-level consequences, enabling me to identify engineering and policy interventions that are robust to both realistic economic behavior and the physical constraints of the power grid.

Modeling Next-Generation Large Loads

Macroscopic system studies traditionally represent electricity-demand resources through black-box or operational-boundary models. This abstraction becomes increasingly limiting for emerging large electricity loads. In Bitcoin mining facilities, AI data centers, and similar intensive resources, grid-facing demand is not simply an exogenous profile; it is the output of internal operational and economic dynamics that evolve over time and determine how much electricity the resource actually consumes under changing system conditions.

Internal operational states define grid-facing variability in AI data centers. In my work on AI data centers¹, I demonstrate that colocating batch and inference workloads causes aggregate power variability and short-horizon ramping to decouple at intermediate workload compositions. Consequently, hybrid systems can appear smoother in aggregate than either batch-only or inference-only configurations, yet they remain difficult for the grid to follow over short timescales.

Using a trace-calibrated shared-GPU framework, I identify the underlying computational mechanism behind this decoupling. By systematically varying batch and inference workload shares while holding overall GPU utilization fixed, I map how internal scheduling creates grid-facing impacts: Inference workloads are latency-sensitive and receive priority access to GPUs; thus, their demand fluctuations pass directly into realized power draw. Batch workloads are delay-tolerant, acting as a buffer that absorbs unused GPU capacity when inference demand recedes. At intermediate workload mixes, this buffering reduces aggregate GPU-utilization variability. However, it fails to eliminate short-horizon ramping because inference-driven spikes are only partially offset by batch dynamics. Furthermore, I find that the magnitude and shape of this variability-ramping decoupling depend heavily on internal operating states, including total system utilization and workload-specific serving parameters. High utilization makes spare GPU capacity scarce, undermining the smoothing role of batch workloads and accelerating the onset of rapid ramping as inference shares rise. Additionally, batch flexibility dictates the strength of this buffer: jobs that can seamlessly pause and resume respond effectively to changing residual capacity, whereas rigid batch structures weaken the buffering effect and shift where ramping peaks.

Therefore, AI data-center impacts on power grids cannot be characterized by annual energy metrics or stylized hourly profiles alone; they are a direct function of the internal operating states that drive real-time demand.

¹S. Majumder, M. Yu, and L. Xie. “Workload composition smooths aggregate power demand while sustaining short-horizon ramps in AI data centers.” arXiv preprint arXiv:2604.10769, 2026.

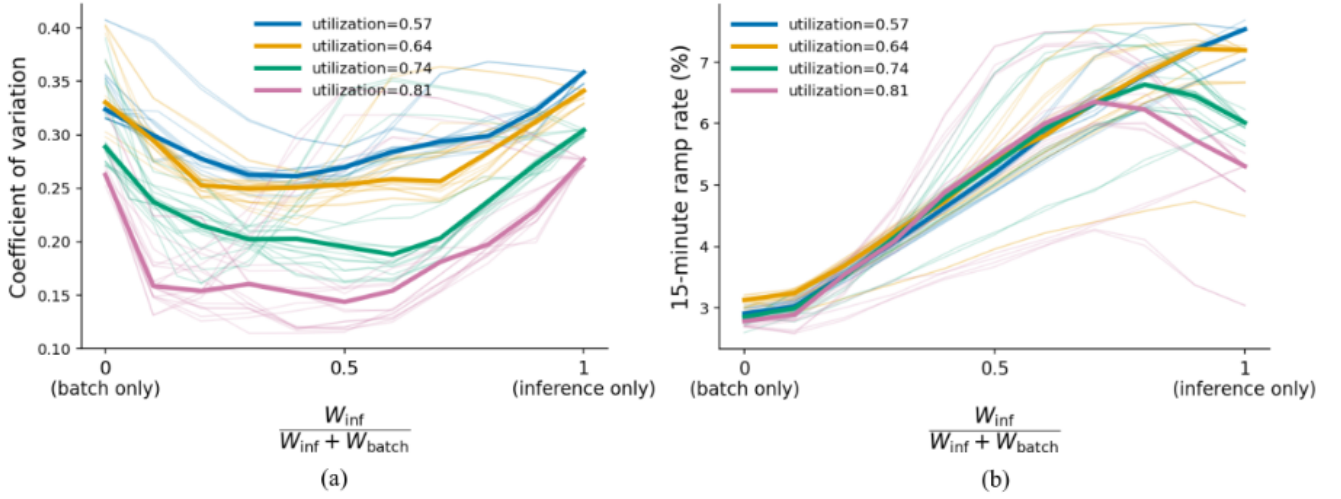


Figure 1: Effect of workload composition on AI data-center power variability and ramping. The horizontal axis shows the inference workload share, $W_{\text{inf}}/(W_{\text{inf}} + W_{\text{batch}})$, ranging from batch-only operation to inference-only operation. Thick lines show average trends at different utilization levels.

Internal economic states determine flexibility of Bitcoin-mining demand. A related challenge arises when modeling Bitcoin-mining loads. While many top-down aggregate models indicate that short-run demand response is small, heterogeneous, and highly concentrated in specific hours, they fail to explain the underlying economic mechanisms driving that response. In an earlier work², I observed similar time-varying price responsiveness in Bitcoin-mining loads; my recent research³ explains this behavior using causal-inference tools.

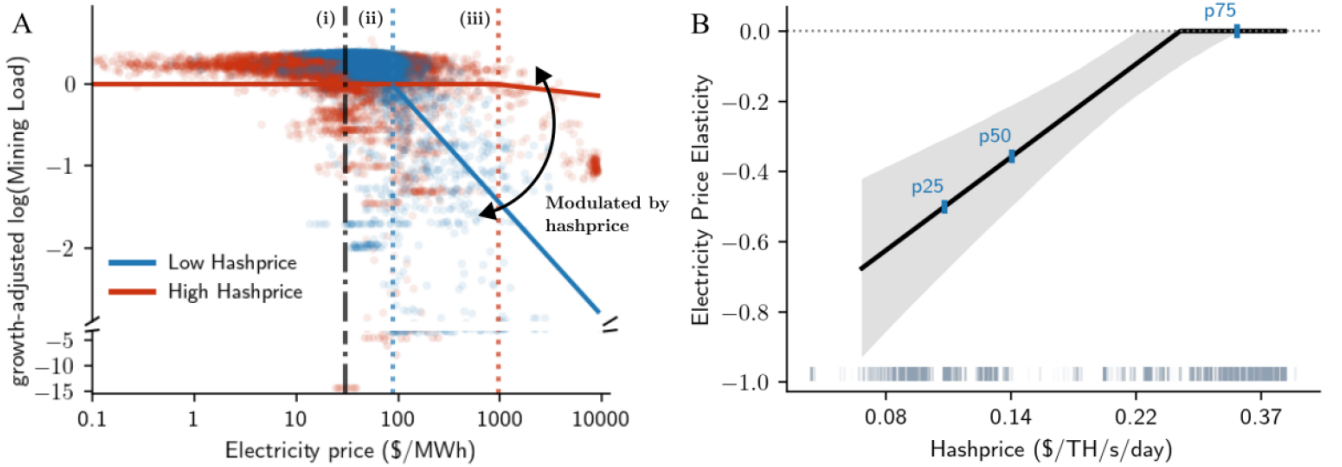


Figure 2: Hashprice dependence of Bitcoin-mining demand response. Mining load becomes more responsive to electricity prices under low hashprice, while high hashprice weakens curtailment even at elevated electricity prices.

The key finding is that Bitcoin-mining load flexibility is governed by internal economic thresholds within mining facilities. Although individual device-level decisions remain unobserved, the aggregate demand response is consistent with facilities operating heterogeneous hardware arrays around distinct breakeven points. As electricity-sector costs rise, an increasing share of less-efficient mining capacity curtails. Conversely, when the

²S. Majumder, I. Aravena, and L. Xie. “An econometric analysis of large flexible cryptocurrency-mining consumers in electricity markets.” In: *Proceedings of the 58th Hawaii International Conference on System Sciences (HICSS)*. Waikoloa Village, HI, USA, 2025.

³S. Majumder. “Hashprice moderates the electricity demand response of Bitcoin miners.” arXiv preprint arXiv:2606.00587v2, 2026.

global “hashprice” (a metric of expected per unit mining revenue) rises, a wider band of hardware remains profitable at any given electricity cost, significantly weakening the aggregate demand response. I demonstrate this behavioral mechanism across two distinct electricity-sector cost channels: wholesale electricity prices and coincident-peak-based transmission charges. In both cases, price responsiveness degrades when hashprice is high. This threshold shift is most visible under wholesale price exposure: higher hashprices push the implied curtailment threshold to significantly higher wholesale price points, showing how favorable external revenue conditions force the grid to pay a premium to trigger demand response.

Ultimately, these findings show that while Bitcoin mining facilities represent a massive theoretical source of demand response, the actual flexibility available to grid operators is highly dependent on economic conditions completely external to the electricity sector.

Planning and Cost-Allocation Frameworks for Modern Power Systems

Macroscopic planning studies traditionally evaluate power-grid infrastructure through the lens of aggregate welfare. This abstraction is essential for large-scale planning, but infrastructure benefits are rarely uniform, and system-level welfare metrics do not by themselves identify how costs should be distributed across stakeholders. My earlier work in transmission expansion and power-quality investment touched upon planning and cost-allocation frameworks that address these linked physical and institutional challenges.

In transmission planning, I found that although least-cost generation and transmission expansion can significantly reduce overall generation costs, it may create localized congestion and increase transmission losses if the network is not expanded appropriately. To ensure that least-cost planning does not come at the expense of network performance, I developed a two-stage transmission planning framework⁴. The first stage minimizes the combined cost of generation and transmission expansion, treating electricity demand as reliably served. If a blended system-performance metric consisting of congestion, electricity cost, network losses, and voltage profile deteriorates under that plan, the second stage incorporates congestion-cost minimization into the main planning objective.

The transmission-planning framework identifies where network expansion is valuable. But it does not answer how the cost of a shared grid investment should be divided among the users who benefit from it. I studied this question in the context of voltage-sag mitigation using dynamic voltage restorers in distribution systems⁵. Industrial customers can be highly sensitive to voltage sags, but their willingness to pay for mitigation varies widely. Dynamic voltage restorers exhibit economies of scale because a single device can protect multiple customers, often at lower total cost than separate customer-level solutions. Since some customers may benefit from shared mitigation while preferring not to contribute in proportion to that benefit, this creates a free-riding problem. I therefore modeled voltage-sag mitigation as a common-pool-resource allocation problem.

In distribution networks, customers are separated by feeder topology. Therefore, feasible customer coalitions are constrained by radial distribution-network structure, making the common-pool resource only partially excludable. I developed a graph-partitioning approach to identify technically viable customer groups and combined it with cooperative-game-theoretic allocation rules to distribute the value created by the mitigation solution. Here, stable customer coalitions can form when each customer receives at least as much utility as it could obtain by defecting to form an alternative subgroup. In the studied cases, the core of the cooperative game remained non-empty, showing that shared mitigation investments could be sustained through voluntary participation.

Algorithms for Autonomous Power Grid Operation

Rapidly changing power-system conditions require control tools that can assist grid operators in real-time decision-making. This challenge arises in both distribution systems with thousands of distributed energy resources and transmission systems exposed to fast-moving external hazards. Through two examples, I demon-

⁴**S. Majumder**, R. M. Shereef, and S. A. Khaparde. “Two-stage algorithm for efficient transmission expansion planning with renewable energy resources.” *IET Renewable Power Generation*, vol. 11, no. 3, pp. 320–329, 2017.

⁵**S. Majumder**, S. A. Khaparde, A. P. Agalgaonkar, S. V. Kulkarni, and S. Perera. “Graph theory based voltage sag mitigation cluster formation utilizing dynamic voltage restorers in radial distribution networks.” *IEEE Transactions on Power Delivery*, vol. 37, no. 1, pp. 18–28, 2022.

strate my contributions to autonomous power-grid operation.

In distribution grids hosting large numbers of edge devices, such as smart solar inverters, residential batteries, and electric vehicles, centralized control schemes scale poorly. I developed a distributed, feedback-based controller that regulates distribution-feeder voltages using only local voltage measurements and peer-to-peer communication with immediate neighbors⁶. In this framework, each edge controller iteratively adjusts its real and reactive power set points to keep distribution-grid voltages within nominal bounds. Through these localized updates, the system converges to an optimal global operating point without requiring centralized observability. I stress-tested the controller in a cyber-physical testbed under realistic communication conditions and verified that it maintains reliable convergence across representative operating scenarios.

A second control challenge arises during extreme weather events, such as wildfires. As the fire front evolves, vulnerable transmission lines and substations may need to be proactively de-energized, causing the network topology to change dynamically. Under these changing system conditions, traditional multi-period optimization models can be difficult to use for real-time operations. Conversely, purely data-driven, learning-based controllers can generate control actions quickly, but they often cannot guarantee feasibility with respect to power-grid physics.

To bridge this operational gap, I developed a hybrid reinforcement-learning-based decision-support tool⁷. Within this framework, a deep deterministic policy gradient controller provides forward-looking guidance based on a compact ‘fire-distance’ metric⁸. The guidance includes proactive transmission-line de-energization, generator set points, and load-shedding decisions. A reactive, single-step, power-flow-constrained optimization layer then adjusts the agent’s recommended generator set points and load-shedding decisions to enforce feasible power-grid operation. I trained and evaluated this controller within an integrated wildfire–power-grid simulation testbed. Across evaluation episodes, the controller consistently reduced load shedding relative to a myopic dispatch benchmark while maintaining runtimes compatible with real-time operation. The learned policy was also robust to wildfire-model mismatch. Even when wildfire propagation speeds were accelerated or slowed relative to the training scenarios, the controller maintained its performance without requiring retraining.

Research Directions

Full Research Statement is available upon request.

⁶P. S. Sarker, **S. Majumder**, M. F. Rafy, and A. K. Srivastava. “Impact analysis of cyber-events on distributed voltage control with active power curtailment.” In: *Proceedings of the 2022 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*. Jaipur, India, 2022.

⁷S. U. Kadir, **S. Majumder**, A. K. Srivastava, A. Chhokra, A. Dubey, H. Neema, and A. Laszka. “Reinforcement learning based proactive control for enabling power grid resilience to wildfire.” *IEEE Transactions on Industrial Informatics*, vol. 20, no. 1, pp. 795–805, 2024.

⁸This metric maps the proximity of the active wildfire front to grid infrastructure.